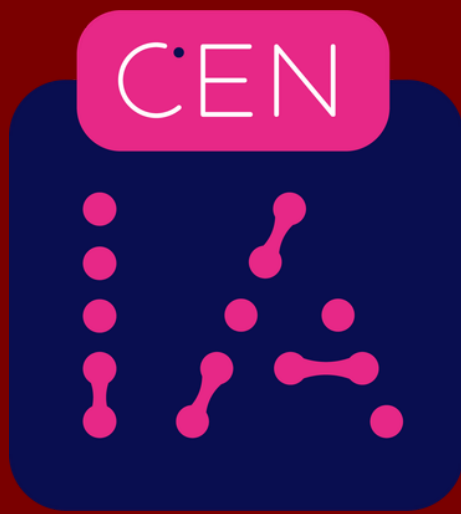




FROM LEGISLATIVE SPEECH TO DISCOURSE NETWORKS

LLM-ASSISTED MEASUREMENT OF IDEATIONAL ROBUSTNESS

IN CHILE’S PENSION REFORM



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Research problem & question

Chile’s individual-account pension system survived repeated legitimacy crises and reform attempts. This study examines how competing justice claims structured the 2022–2025 legislative debate and whether the ideas legitimating individual capitalization remained robust by **persisting, diffusing, and adapting**.
Research question: How did claims preserving or challenging individual capitalization form discourse coalitions, how did these coalitions legitimate their positions, and did preservation ideas persist, diffuse, and adapt during the legislative process?

From justice claims to ideational robustness

Justice principles: Market justice links benefits to contribution, effort, and individual responsibility. Political justice emphasizes need, equality, collective risk, and public intervention [1]. These are interpretive lenses, not network nodes.
Observed concepts: Statements are coded using CARIN deservingness criteria—Control, Attitude, Reciprocity, Identity, and Need [2]—and pension-specific concepts: individual ownership, individual capitalization, solidarity, pension adequacy, and financial sustainability. Agreement or disagreement with these concepts forms the discourse network.
Legitimation: Moralization, rationalization, narrativization, authorization, and normalization capture how actors present their positions as acceptable [3,4]. They are coded separately from the concepts.
Ideational robustness: Robust ideas maintain their core orientation while adapting peripheral elements [5]. Robustness is assessed as permanence, diffusion, and adaptation.

Analytical strategy

- **H1 — Discourse coalitions:** Actor–concept positions are projected into a signed actor network. Positive ties represent shared concept–stance positions; negative ties represent conflict. Signed spin-glass community detection identifies empirical coalitions [6].
- **Q2 — Legitimation:** A cross-classified multilevel multinomial model estimates the primary strategy by coalition, party, actor type, and legislative stage, with random intercepts for actor and session [7].
- **H3a — Permanence:** Coverage, support–opposition balance, and internal connectivity of a fixed preservation core.
- **H3b — Diffusion:** Later adoption of that core by actors initially oriented toward transformation.
- **H3c — Adaptation:** Within-intervention coupling of solidarity with contributory, targeted, or financial constraints.
- **Substantive audit:** The statements supporting the main quantitative patterns are read qualitatively to rule out polysemy, strategic appropriation, and spurious textual proximity.

Current stage & next steps

- Continue with the data processing: video scraping and ASR pipeline.
- Start with annotation experiments with different LLMs.
- Consolidate a prompt and a codebook.
- Run first inference models and build indices.

Corpus, units & LLM-assisted coding

- **Corpus:** 92 documented sessions from the Chamber of Deputies and the Senate, covering plenary and technical committee debate across three deliberative stages of Law No. 21.735.
- Public legislative records are scraped and standardized with a reusable Python pipeline. Video-only sessions are processed using automatic speech recognition, speaker diarization, and semi-automated speaker identification.
- **Unit hierarchy:** legislative stage → session → intervention → DNA statement.
- Each statement links one actor, one concept, and one stance. It may also contain explicit legitimation and one primary plus one optional secondary strategy.
- A researcher-defined codebook guides the LLM. The model returns structured labels, verbatim evidence, and ambiguity flags; it cannot create new categories during production coding.
- Ambiguous cases are manually reviewed. Performance is evaluated by task on held-out human coding using precision, recall, F1, macro-F1, Cohen’s kappa, and confusion matrices.

From statements to discourse coalitions

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